

MTH 310/520 Graph Theory: Quiz 1

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Total: 10 points

Given a graph G , an *ear decomposition* of a graph G is a collection of paths $P_1, \dots, P_m \subseteq G$ and a cycle $P_0 \subseteq G$, such that $\cup_{i=0}^m P_i = G$, and for any $1 \leq i \leq m$, the end points of P_i lie in $V(\cup_{j=0}^{i-1} P_j)$ and no interior point of P_i lies in $V(\cup_{j=0}^{i-1} P_j)$. The cycle and paths $P_0, \dots, P_m$ are referred to as *ears*. In this quiz, we shall prove the following result that $\kappa(G) \geq 2 \iff G$ admits an ear decomposition.

1) [CO1,CO3] In this question, we show that $\kappa(G) \geq 2 \implies G$ admits an ear decomposition.

a) Show that $\kappa(G) \geq 2$ implies G has a cycle.

b) Define P_0 to be a cycle obtained in Q1.a. Now assume that you have already constructed the ears $P_0, P_1, \dots, P_i$. Define $A := V(\cup_{j=0}^i P_j)$, and $B := V(G) \setminus A$. If $A = V(G)$ does $P_0, \dots, P_i$ necessarily constitute an ear decomposition? If not, show that one can add more ears to complete the ear decomposition.

c) Now, suppose $A \subsetneq V(G)$. Show that you can always add one more ear P_{i+1} .

[Hint: Observe that $B \neq \emptyset$. Can you suitably use Menger's theorem between A and some component of $G[B]$ to construct P_{i+1} ?]

[1+2+4=7 points]

2) [CO1,CO3] In this question, we show the reverse direction, i.e., if G admits an ear decomposition then $\kappa(G) \geq 2$.

a) Let $P_0, \dots, P_m$ be an ear decomposition for G . Then for any $0 \leq i \leq m$, show that $G[V(\cup_{j=0}^i P_j)]$ is connected.

b) For any $v \in V(G)$, define $\text{ear}(v)$ to be the least index i such that $v \in V(P_i)$. Show using Q2.a) that vertices in $V(\cup_{i=0}^{\text{ear}(v)} P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$.

c) Next, show that for any $k > \text{ear}(v)$ vertices in $V(\cup_{i=0}^k P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$ provided vertices in $V(\cup_{i=0}^{k-1} P_i) \setminus \{v\}$ lie in the same component of $G \setminus \{v\}$.

d) Argue using the above that $\kappa(G) \geq 2$.

[1+1+0.5+0.5=3 points]